The Printed Gaussian Coefficient-Sum Sequence Is Not Eventually Periodic, Even Modulo Three

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AI-assisted disproof of conjecture 00000007686

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Abstract

Both source languages define $p_q(n)$ as the coefficient sum of the Gaussian polynomial $\begin{bmatrix} 2n \\ n \end{bmatrix}_q$. That sum is $c(n) = \binom{2n}{n}$, not the ordinary partition number. The natural-valued sequence is strictly increasing. More strongly, $c(n) \bmod 3$ is not eventually periodic: it has arbitrarily long zero intervals immediately before powers of three, where its value is two. This refutes the source's necessary eventual-periodicity clause at $k = 1$, including the plausible implicit coefficient modulus $p = 3, a = 1$. The formalization uses actual polynomials, a generated semiring congruence, an actual polynomial-ring quotient over $\mathbb{F}_3$, and universal Lucas/zero-interval proofs. No claim about ordinary partition periodicity or a particular unpublished period formula is made.

1 Exact Scope and the Necessary Clause

The complete frozen bilingual source is retained separately. Its identity is:

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Commit:  b355e42906e6eaf52bc7e9d005e0ef02d8aa7400
Blob:  337d3e5e4d97e64cec5d370fa8a3b157ebd77c6f.
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Both languages first mention ordinary partition numbers $p(n)$ modulo p^a, but then *explicitly define* $p_q(n)$ as the coefficient sum of $\begin{bmatrix} 2n \\ n \end{bmatrix}_q$. They assert eventual periodicity in the displayed quotient $\mathbb{N}[q]/(q-1)^k$, an explicit minimal-period formula, and a further classical-period comparison. For a sequence s in any fixed quotient, the necessary clause being refuted is precisely

$$\mathrm{EP}(s): \quad \exists T \in \mathbb{N},\ T > 0,\ \exists N \in \mathbb{N},\ \forall n \geq N, \quad s(n+T) = s(n).$$

A homomorphism carries this property to its image with the same N, T. It is enough to disprove one such necessary image clause at $k = 1$; neither of the additional period-formula assertions is independently evaluated. The parameter witness is $p = 3, a = 1, k = 1$, with a prime and positive exponents. This is not a claim of nonperiodicity for every coefficient modulus: $c(n) \bmod 2$ is zero for $n > 0$.

2 The Actual Gaussian Polynomial and Its Coefficient Sum

Use the standard nonnegative Gaussian polynomial $G(N, K; q) \in \mathbb{N}[q]$, characterized by $G(0, 0; q) = 1$, $G(0, K; q) = 0$ for $K > 0$, and

$$\begin{aligned} G(N+1, 0; q) &= 1, \\ G(N+1, K+1; q) &= G(N, K; q) + q^{K+1} G(N, K+1; q). \end{aligned}$$

The recursion uniquely determines all entries. It gives $G(N,K;q)=0$ for $K>N$ and $G(N,N;q)=1$. This is an ordinary q-Pascal characterization, not a sequence declared equal to the desired scalar. For comparison with the product convention, in $\mathbb{Q}(q)$ set

$$H(N,K)=\prod_{j=1}^{K}\frac{1-q^{N-K+j}}{1-q^{j}}\quad(0\le K\le N).$$

For $0<K<N$, the ratios to $H(N,K)$ of $H(N-1,K-1)$ and $H(N-1,K)$ are $(1-q^K)/(1-q^N)$ and $(1-q^{N-K})/(1-q^N)$. The first plus q^K times the second equals one. Together with the boundary values, this verifies the same recursion and hence the standard identification. The Lean definition uses the recursive polynomial characterization; a separate rational-function product theorem is not advertised as formalized.

For any polynomial f, evaluation at one is its finite coefficient sum:

$$\sum_j [q^j]f=f(1).$$

Evaluating q-Pascal at one gives ordinary Pascal recursion and the same boundary values. Induction in N, simultaneously for all K, proves

$$G(N,K;1)=\binom{N}{K},\qquad c(n):=\sum_j [q^j]G(2n,n;q)=\binom{2n}{n}.$$

There is no evaluation of a 0/0 product at $q=1$. Both the actual polynomial recursion and the finite coefficient-sum/evaluation bridge are proved in Lean. The strict monotonicity of actual central binomial coefficients also proves $c(n+T)>c(n)$ for every $T>0$, excluding natural-valued eventual periodicity. This monotonicity is *not* used to infer modular nonperiodicity.

3 The q=1 Quotients Are Not Substituted Models

Natural-coefficient polynomials form a semiring, not a ring. Thus the printed $\mathbb{N}[q]/(q-1)$ is not silently fed to a commutative-ring ideal API. The standard semiring presentation is the least additive and multiplicative congruence generated by $q\sim 1$. We prove it equals the kernel congruence of the actual evaluation homomorphism $E_1:\mathbb{N}[q]\to\mathbb{N}$.

Indeed, $E_1(q)=E_1(1)$, so every generated relation preserves evaluation. Conversely $q\sim 1$ implies $q^j\sim 1$. Expanding any polynomial as its finite sum of monomials gives $f\sim C(f(1))$. Equal evaluations therefore give congruent polynomials. Constants retain their natural values, so the source-defined scalar is represented by $C(c(n))$ and cannot acquire a period in this quotient.

To cover the background's plausible omitted coefficient modulus, use the genuine ring quotient

$$R=\mathbb{F}_3[q]/\langle q-1\rangle.$$

Its standard evaluation isomorphism $R\simeq\mathbb{F}_3$ sends $C(c(n))$ to $c(n)$ mod 3. The Lean theorem proves equality of the two actual quotient classes *if and only if* the corresponding natural remainders agree. Reduction of the actual polynomial $G(2n,n;q)$ also commutes with evaluation and has the same class as $C(c(n))$. Thus retaining the Gaussian polynomial instead of its source-defined coefficient sum does not rescue periodicity. Reading the printed natural quotient through the ring completion $\mathbb{Z}[q]/\langle q-1\rangle$ has the same necessary reduction to $\mathbb{F}_3$; no separate Lean ring-completion theorem is claimed or needed.

These are the conventional q=1 meanings, not an assertion that the literal natural-semiring ideal notation is well-formed. Replacing algebraic subtraction by truncated natural subtraction would impose q=0, a different repair. Replacing $c(n)$ by ordinary $p(n)$ would change the explicit definition in both languages. Neither repair is performed here.

4 Universal Nonperiodicity Modulo Three

Write $d(n) = c(n) \bmod 3$. One-digit Lucas congruences give, for every $m \geq 0$,

$$d(3m) = d(m), \qquad d(3m+1) = 2d(m) \pmod 3, \qquad d(3m+2) = 0. \tag{1}$$

For clarity, the last formula does not assume carry-free doubling: $2(3m+2) = 3(2m+1)+1$, whereas $3m+2$ has low ternary digit two; the low-digit binomial factor is $\binom{1}{2} = 0$. The first two formulas similarly have low factors $\binom{0}{0} = 1$ and $\binom{2}{1} = 2$. The formal proof applies the actual Mathlib Lucas theorem.

Lemma 1. *For every $m \geq 0$, $d(3^m) = 2$ and*

$$d(n) = 0 \quad \textit{whenever} \quad 2 \cdot 3^m \leq n < 3^{m+1}.$$

Proof. The power statement follows from $d(1) = 2$ and the first formula in (1). The zero-interval base case is $n = 2$. For the induction step, divide an index in the next interval by three. Its quotient belongs to the preceding zero interval; its remainder is zero, one or two. The three formulas in (1) then give zero in every case. This proves the whole interval, not a bounded sample. □

Theorem 1. *The sequence $\binom{2n}{n} \bmod 3$ is not eventually periodic.*

Proof. Let any proposed period $T > 0$ and starting point N be given. Choose m with $A = 3^m \geq \max(T, N)$ and put $n = 3A - T$. Then $2A \leq n < 3A$ and $n \geq N$. Lemma 1 yields

$$d(n) = 0, \qquad d(n+T) = d(3^{m+1}) = 2,$$

contradicting the proposed period. In Lean the choice is explicitly $m = \max(T, N)$, using $m < 3^m$; there is no existence assumption hiding the obstruction. All T, N are quantified. □

Composing this theorem with the actual coefficient-sum and quotient bridges negates the shared necessary clause at $p = 3, a = 1, k = 1$. Recursive digit structure is compatible with this nonperiodicity and is not an eventual fixed additive period. We do not exchange the quantifiers of coefficient-by-coefficient periods of an unquotiented infinite family.

5 Formal Scope, Reproduction and Disclosure

The four mathematical modules are Gaussian, QOnePresentation, ModThree and PeriodicityNegation. They export 33 public theorems, with all compiled types and individual axiom footprints audited. The final theorem `source_k_one_not_periodic` gives the natural-presentation and actual modulo-three quotient negations. A further theorem covers the coefficient-reduced Gaussian polynomial itself. No forbidden proof escapes or additional mathematical axioms are used.

Lean is pinned to 4.33.1 and Mathlib to

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0df444a360eaa60ab8c11dca51a86af692955474.
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Only `propext`, `Classical.choice` and `Quot.sound` occur. Finite exact-integer diagnostics are regression evidence, not the universal proof. Installed Lean and fixed dependency build caches remain trust inputs; no full-dependency clean build, empty-machine bootstrap, cross-platform run or hosted-CI result is implied. Independent AI review is not human expert endorsement or organizer acceptance. Publication, priority, scores and awards are not claimed by this mathematical report.